

5-Mark Questions & Answers — DBMS Module 6: Foundations of IBM Db2

Q1. Explain IBM Db2 overview, features (BLU Acceleration, pureScale, HADR), and industry use cases.

- **BLU Acceleration:** In-memory columnar storage and SIMD vector processing for fast queries.
- **pureScale:** Active-active clustering for 24/7 continuous availability.
- **HADR:** High Availability Disaster Recovery standby database replication.
- **Use Cases:** Core Banking, Healthcare Claim Systems, Large-scale E-commerce Analytics.

Q2. Write Db2 SQL statements for Table Creation, Constraints (PK, FK, CHECK), Data Manipulation, NULL handling (VALUE()), and JOINS.

```
-- DDL
CREATE TABLE DEPT (DEPT_ID INT PRIMARY KEY, DNAME VARCHAR(50));
CREATE TABLE EMP (
    EMP_ID INT PRIMARY KEY,
    NAME VARCHAR(50) NOT NULL,
    SALARY DECIMAL(10,2) CHECK (SALARY > 0),
    DEPT_ID INT REFERENCES DEPT(DEPT_ID)
);

-- NULL Handling
SELECT NAME, VALUE(SALARY, 0.00) AS NET_SALARY FROM EMP;

-- Joins
SELECT E.NAME, D.DNAME FROM EMP E INNER JOIN DEPT D ON E.DEPT_ID = D.DEPT_ID;
```

Q3. Explain Db2 Backup and Recovery utilities (db2 BACKUP, RESTORE, ROLLFORWARD).

```
# Offline Backup
db2 BACKUP DATABASE STUDENTDB TO "C:\Backups"

# Online Backup with Logs
db2 BACKUP DATABASE STUDENTDB ONLINE TO "C:\Backups" INCLUDE LOGS

# Restore & Rollforward
db2 RESTORE DATABASE STUDENTDB FROM "C:\Backups"
db2 ROLLFORWARD DATABASE STUDENTDB TO END OF LOGS AND COMPLETE
```

Q4. Explain Db2 Indexing, B+ Tree performance tuning, and Db2 Warehouse on Cloud.

- **B+ Tree Indexing:** `CREATE UNIQUE INDEX IX_EMP ON EMP(EMAIL);`
- **Tuning:** Use `RUNSTATS` after bulk inserts and `REORG` to clear table fragmentation.

- **Db2 Warehouse:** Scalable cloud analytical database using column storage and parallel execution.